

Dynamic Greetings Component

1. Greeting.js Component

jsx

```
import React, { useState, useEffect } from 'react';

const Greeting = () => {
  const [currentTime, setCurrentTime] = useState(new Date());
  const [greeting, setGreeting] = useState('');
  const [icon, setIcon] = useState('');
  const [message, setMessage] = useState('');

  // Function to determine greeting based on hour
  const getGreeting = (hour) => {
    if (hour < 12) {
      return {
        text: 'Good Morning',
        icon: '☀️',
        message: 'Start your day with a smile!'
      };
    } else if (hour >= 12 && hour < 18) {
      return {
        text: 'Good Afternoon',
        icon: '🌞',
        message: 'Keep up the great work!'
      };
    } else {
      return {
        text: 'Good Evening',
        icon: '🌙',
        message: 'Time to relax and unwind.'
      };
    }
  };

  // Update greeting when time changes
  useEffect(() => {
    const hour = currentTime.getHours();
    const newGreeting = getGreeting(hour);
    setGreeting(newGreeting.text);
    setIcon(newGreeting.icon);
    setMessage(newGreeting.message);

    // Optional: Update time every minute
    const timer = setInterval(() => {
      setCurrentTime(new Date());
    }, 60000);

    // Cleanup interval on component unmount
    return () => clearInterval(timer);
  }, [currentTime]);

  // Format time for display
  const formatTime = (date) => {
    return date.toLocaleTimeString('en-US', {
      hour: '2-digit',
      minute: '2-digit',
      second: '2-digit',
      hour12: true
    });
  };

  // Format date for display
  const formatDate = (date) => {
    return date.toLocaleDateString('en-US', {
```

```

    weekday: 'long',
    year: 'numeric',
    month: 'long',
    day: 'numeric'
  });
};

// Get background color based on time of day
const getBackgroundColor = () => {
  const hour = currentTime.getHours();
  if (hour < 12) return 'linear-gradient(135deg, #f6d365 0%, #fda085 100%)';
  if (hour < 18) return 'linear-gradient(135deg, #fbc2eb 0%, #a6c1ee 100%)';
  return 'linear-gradient(135deg, #667eea 0%, #764ba2 100%)';
};

// Inline styles
const styles = {
  container: {
    display: 'flex',
    justifyContent: 'center',
    alignItems: 'center',
    minHeight: '100vh',
    background: getBackgroundColor(),
    fontFamily: 'Arial, sans-serif',
    transition: 'background 1s ease',
    padding: '20px'
  },
  card: {
    backgroundColor: 'rgba(255, 255, 255, 0.9)',
    borderRadius: '20px',
    padding: '40px',
    boxShadow: '0 10px 40px rgba(0, 0, 0, 0.2)',
    textAlign: 'center',
    maxWidth: '500px',
    width: '100%',
    backdropFilter: 'blur(10px)',
    border: '1px solid rgba(255, 255, 255, 0.3)'
  },
  icon: {
    fontSize: '80px',
    marginBottom: '20px',
    animation: 'float 3s ease-in-out infinite'
  },
  greeting: {
    fontSize: '48px',
    fontWeight: 'bold',
    color: '#333',
    margin: '20px 0',
    textShadow: '2px 2px 4px rgba(0,0,0,0.1)'
  },
  time: {
    fontSize: '36px',
    fontWeight: '300',
    color: '#555',
    margin: '10px 0',
    fontFamily: 'monospace'
  },
  date: {
    fontSize: '20px',
    color: '#666',
    margin: '10px 0',
    fontWeight: '500'
  },
  message: {
    fontSize: '24px',
    color: '#444',
    margin: '20px 0',
    fontStyle: 'italic',
    padding: '15px',
    backgroundColor: 'rgba(255, 255, 255, 0.5)',
    borderRadius: '10px'
  }
};

```

```

    },
    timeBadge: {
      display: 'inline-block',
      padding: '8px 16px',
      backgroundColor: '#007bff',
      color: 'white',
      borderRadius: '20px',
      fontSize: '16px',
      marginTop: '20px'
    }
  }
};

// Add keyframes animation
const styleSheet = document.createElement('style');
styleSheet.textContent = `
  @keyframes float {
    0% { transform: translateY(0px); }
    50% { transform: translateY(-20px); }
    100% { transform: translateY(0px); }
  }
`;
document.head.appendChild(styleSheet);

return (
  <div style={styles.container}>
    <div style={styles.card}>
      <div style={styles.icon}>{icon}</div>
      <h1 style={styles.greeting}>{greeting}</h1>
      <div style={styles.time}>{formatTime(currentTime)}</div>
      <div style={styles.date}>{formatDate(currentTime)}</div>
      <div style={styles.message}>{message}</div>
      <div style={styles.timeBadge}>
        Hour: {currentTime.getHours()}: {String(currentTime.getMinutes()).padStart(2, '0')}
      </div>
    </div>
  </div>
);
};

export default Greeting;

```

2. Alternative Simpler Version (GreetingSimple.js)

```

jsx

import React, { useState, useEffect } from 'react';

const GreetingSimple = () => {
  const [greeting, setGreeting] = useState('');
  const [hour, setHour] = useState(new Date().getHours());

  useEffect(() => {
    // Update hour every minute
    const interval = setInterval(() => {
      setHour(new Date().getHours());
    }, 60000);

    return () => clearInterval(interval);
  }, []);

  // Conditional rendering based on hour
  const renderGreeting = () => {
    if (hour < 12) {
      return (
        <div style={stylesSimple.morning}>
          <h1> Good Morning</h1>
          <p>Current Hour: {hour}</p>
        </div>
      );
    } else if (hour >= 12 && hour < 18) {

```

```

    return (
      <div style={stylesSimple.afternoon}>
        <h1>☀ Good Afternoon</h1>
        <p>Current Hour: {hour}</p>
      </div>
    );
  } else {
    return (
      <div style={stylesSimple.evening}>
        <h1>    Good Evening</h1>
        <p>Current Hour: {hour}</p>
      </div>
    );
  }
};

const stylesSimple = {
  container: {
    display: 'flex',
    justifyContent: 'center',
    alignItems: 'center',
    height: '100vh',
    fontFamily: 'Arial, sans-serif'
  },
  morning: {
    backgroundColor: '#FFE87C',
    padding: '40px',
    borderRadius: '10px',
    color: '#333'
  },
  afternoon: {
    backgroundColor: '#FFB347',
    padding: '40px',
    borderRadius: '10px',
    color: '#333'
  },
  evening: {
    backgroundColor: '#4A4E69',
    padding: '40px',
    borderRadius: '10px',
    color: 'white'
  }
};

return (
  <div style={stylesSimple.container}>
    {renderGreeting()}
  </div>
);
};

export default GreetingSimple;

```

3. App.js - Main Application Component

jsx

```

import React from 'react';
import Greeting from './Greeting';
// import GreetingSimple from './GreetingSimple'; // Alternative version

function App() {
  return (
    <div className="App">
      <Greeting />
      { /* <GreetingSimple /> */ }
    </div>
  );
}

```

```
export default App;
```

4. index.js - Entry Point

jsx

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import './index.css';
import App from './App';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

5. index.css - Global Styles

CSS

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

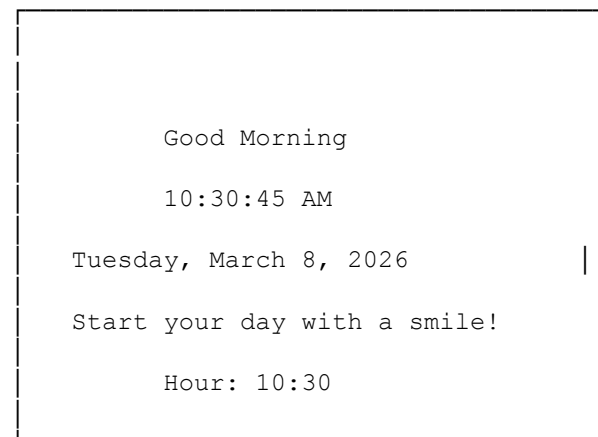
body {
  font-family: -apple-system, BlinkMacSystemFont, 'Segoe UI', 'Roboto', 'Oxygen',
    'Ubuntu', 'Cantarell', 'Fira Sans', 'Droid Sans', 'Helvetica Neue',
    sans-serif;
  -webkit-font-smoothing: antialiased;
  -moz-osx-font-smoothing: grayscale;
  overflow: hidden;
}

code {
  font-family: source-code-pro, Menlo, Monaco, Consolas, 'Courier New',
    monospace;
}
```

Expected Output Based on Time

Morning (Before 12 PM)

text



Afternoon (12 PM - 6 PM)

text



Good Afternoon

2:15:30 PM

Tuesday, March 8, 2026

Keep up the great work!

Hour: 14:15

Evening (After 6 PM)

text



Good Evening

8:45:20 PM

Tuesday, March 8, 2026

Time to relax and unwind.

Hour: 20:45

Key Features Implemented

- ✔ **Conditional Rendering:**
 - Uses if-else conditions based on hour
 - Displays appropriate greeting message
 - Changes icon based on time of day
- ✔ **Time Calculation:**
 - Uses JavaScript Date object
 - Gets current hour (0-23)
 - Real-time updates every minute
- ✔ **Dynamic Styling:**
 - Background gradient changes with time
 - Animated floating icon
 - Responsive card layout
 - Professional color scheme

✓ Additional Features:

- Live time display with seconds
- Full date display
- Motivational messages
- Time badge showing current hour
- Smooth transitions

Time Logic Explained

javascript

```
const getGreeting = (hour) => {  
  if (hour < 12) {  
    return 'Good Morning'; // 12 AM - 11:59 AM  
  } else if (hour >= 12 && hour < 18) {  
    return 'Good Afternoon'; // 12 PM - 5:59 PM  
  } else {  
    return 'Good Evening'; // 6 PM - 11:59 PM  
  }  
};
```

To Run the Application

1. Create a new React app:

bash

```
npx create-react-app greeting-app  
cd greeting-app
```

2. Replace the default files with the code above

3. Start the application:

bash

```
npm start
```

4. Open <http://localhost:3000> to view it

The greeting will automatically update based on your local system time, providing a personalized experience throughout the day!

